Marking Schemes

This document was prepared for markers' reference. It should not be regarded as a set of model answers. Candidates and teachers who were not involved in the marking process are advised to interpret its content with care.

Chemistry
Paper 1

SECTION A

Question No. Part I	Key	Question No. Part II	Key
1.	C (72%)	25.	B (59%)
2.	C (71%)	26.	D (69%)
3.	A (83%)	27.	A (66%)
4.	B (80%)	28.	B (77%)
5.	C (84%)	29.	B (40%)
6.	B (58%)	30.	A (65%)
7.	D (60%)	31.	B (60%)
8.	C (75%)	32.	A (72%)
9.	C (20%)	33.	D (61%)
10.	A (60%)	34.	C (76%)
11.	C (73%)	35.	A (65%)
12.	D (91%)	36.	D (79%)
13.	B (68%)		
14.	B (86%)		
15.	C (66%)		
16.	D (77%)		
17.	A (80%)		
18.	D (19%)		
19.	D (79%)		
20.	C (91%)		
21.	A (68%)		
22.	D (54%)		
23.	A (79%)		
24.	C (69%)		

Note: Figures in brackets indicate the percentages of candidates choosing the correct answers.

General Marking Instructions

1. In order to maintain a uniform standard in marking, markers should adhere to the marking scheme agreed at the markers' meeting.

2. The marking scheme may not exhaust all possible answers for each question. Markers should exercise their professional discretion and judgment in accepting alternative answers that are not in the marking scheme but are correct and well reasoned.

3. In questions asking for a specified number of reasons or examples etc. and a candidate gives more than the required number, the extra answers should not be marked. For instance, in a question asking candidates to provide two examples, and if a candidate gives three answers, only the first two should be marked.

4. In cases where a candidate answers more questions than required, the answers to all questions should be marked. However, the excess answer(s) receiving the lowest score(s) will be disregarded in the calculation of the final mark.

5. Award zero marks for answers which are contradictory.

6. Chemical equations should be balanced except those in reaction schemes for organic synthesis. For energetics, the chemical equations given should include the correct state symbols of the chemical species involved.

7. In the question paper, questions which assess candidates' communication skills are marked with an asterisk (*). For these questions, the mark for effective communication (1 mark per question) will be awarded if candidates can produce answers which are easily understandable. No marks for effective communication will be awarded if the answers produced by candidates contain a lot of irrelevant materials and/or wrong concepts in chemistry.

SECTION B

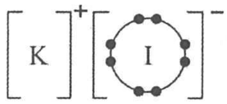
Part I

Marks

1. (a) They are isotopes. 1

(b) 18 1

(c) 1



(d) Hydrogen iodide ionises in water to form mobile ions. 1

(e) Potassium iodide would have a higher melting point because
- melting of potassium iodide needs a lot of energy to break the strong ionic bonds between K^+ ions and I^- ions in a giant ionic structure. 1
- melting of hydrogen iodide only needs little energy to break the weak van der Waals' forces between molecules in a simple molecular structure. 1

2. (a)
- oxygen / O_2 1
- It relights a glowing splint. 1

(b) (i) Relative atomic mass of **X** $= 2.819 \div \left[\frac{(3.028-2.819)}{16} \times 2\right]$ 2
$= 108$

(ii) silver / Ag 1

(c) Yes.
The oxidation number of **X** decreases from +1 to 0. / The oxidation number of O increases from –2 to 0. 1

3. (a) $NaHCO_3(s) + HCl(aq) \rightarrow NaCl(aq) + CO_2(g) + H_2O(l)$ / 1
$NaHCO_3(s) + H^+(aq) \rightarrow Na^+(aq) + CO_2(g) + H_2O(l)$

(b) No. of moles of HCl(aq) used = 0.644 × 0.0252 = 0.01623 2
No. of moles of $NaHCO_3(s)$ in the antacid sample = 0.01623
Percentage by mass of $NaHCO_3(s)$ in the antacid sample
= 84 × 0.01623 ÷ 1.52 × 100% = 89.7%

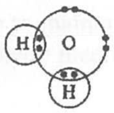
(c) (i)
- methyl orange 1
- From yellow to orange 1

(ii) pH meter / data-logger connected with a pH sensor 1

(d) No gas is given out from the reaction between $Mg(OH)_2(s)$ and stomach acid, while carbon dioxide gas is given out from the reaction between $NaHCO_3(s)$ and stomach acid, this leads to uncomfortable feeling in stomach. 1

Marks

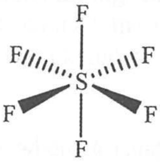
4. (a) (i) 2


+ $[H]^+$

(ii) A lone pair of electrons on oxygen atom in H_2O is donated to H^+ to form a dative covalent bond. 1

(b) No, because boron atom in BF_3 molecule has only 6 outermost shell electrons. 1

(c) (i) 1



(ii) No, because a SF_6 molecule is symmetrical, the polarities of the S—F bonds in SF_6 cancel out each other. 1

(d)
- Both BF_3 and SF_6 are non-polar molecules, there are weak van der Waals' forces between their molecules. 1
- As the molecular size of SF_6 is larger than that of BF_3, the van der Waals' forces between SF_6 molecules are stronger than those between BF_3 molecules. 1
- There are stronger hydrogen bonds between H_2O molecules. 1

5. (a) A primary cell is a cell that cannot be recharged. 1

(b) (i) $H_2(g) + 2OH^-(aq) \rightarrow 2H_2O(l) + 2e^-$ 1

(ii) Hydrogen is flammable. / Concentrated KOH(aq) is corrosive. 1

(c) (i) $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$ 1

(ii) $4Al(s) + 3O_2(g) + 6H_2O(l) \rightarrow 4Al(OH)_3(s)$ 1

(iii) Electrolysis of molten aluminium oxide 1

6. (a) substitution 1

(b) light / diffused sunlight / ultra-violet 1

(c) (i) Chlorine free radical is a species which has one unshared electron. 1

(ii) $Cl\bullet + CH_4 \rightarrow \boxed{\bullet CH_3} + \boxed{HCl}$ 1

$\boxed{\bullet CH_3} + \boxed{Cl\bullet} \rightarrow CH_3Cl$ 1

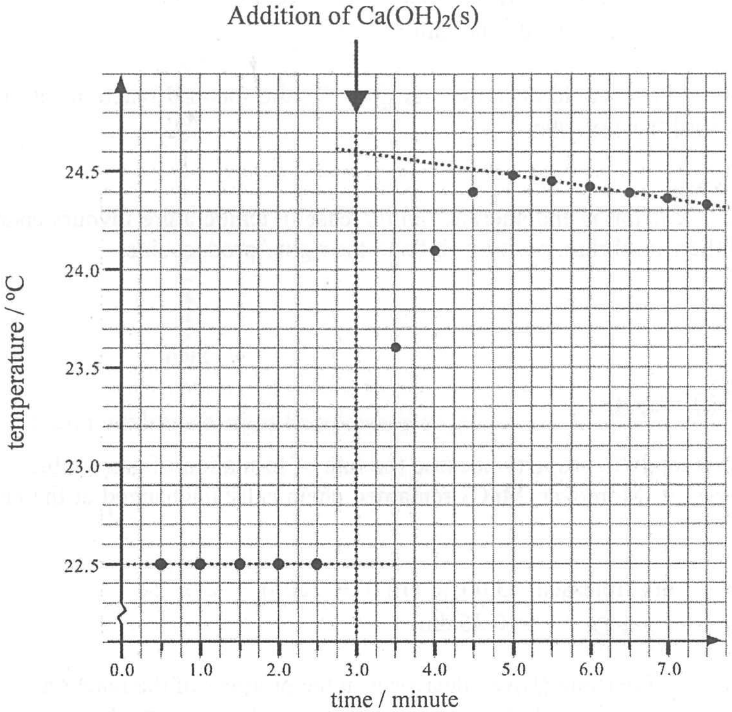
(d) Methane / CH_3Cl undergoes further substitution with chlorine to form CH_2Cl_2 / $CHCl_3$ / CCl_4. 1

(e) 1

Marks

7. (a) $Ca(OH)_2(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + 2H_2O(l)$ / 1
$Ca(OH)_2(s) + 2H^+(aq) \rightarrow Ca^{2+}(aq) + 2H_2O(l)$

(b) (i) 1



The greatest temperature rise = 2.1 °C 1

(ii) Energy released during the reaction = 100.0 × 1.00 × 4.2 × 2.1 = 882 J 3
Enthalpy change of the reaction = –882 ÷ 1000 ÷ (0.502 ÷ 74.1) = –130 kJ mol^{-1}
Enthalpy change of neutralisation = –130 ÷ 2 = –65 kJ mol^{-1}

(c) $\frac{1}{2}Ca(OH)_2(s) + HCl(aq) \rightarrow \frac{1}{2}CaCl_2(aq) + H_2O(l)$ $\Delta H_n^\ominus = -58.6$ kJ mol^{-1} 3
$CaO(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$ $\Delta H_n^\ominus = -186.0$ kJ mol^{-1}
$CaO(s) + H_2O(l) \rightarrow Ca(OH)_2(s)$ $\Delta H^\ominus = y$

$y = -186.0 - (-58.6) \times 2$
$= -68.8$ kJ mol^{-1}

8. Chemical knowledge 5
Similarities:
- Both tin-plating and galvanising involve coating iron with a thin layer of metal.
- The layer prevents iron from contacting water and oxygen.

Differences:
- Tin-plated iron will rust when tin coating is scratched off, but galvanised iron will not rust when zinc coating is scratched off.
- When the tin coating is scratched off, tin-plated iron will rust because tin loses electrons less readily than iron.
- When the zinc coating is scratched off, galvanised iron will not rust because zinc loses electrons more readily than iron.

Communication mark 1

Part II

Marks

9. (a) (i) Reaction quotient $Q_c = [PCl_3(g)][Cl_2(g)] / [PCl_5(g)]$ 3

$= (0.16 \div 4) \times (0.16 \div 4) \div (0.84 \div 4)$

$= 7.6 \times 10^{-3}$ mol dm^{-3}

(ii) Concentration of PCl_5 would decrease. As $Q_c < K_c$, the forward reaction rate is greater than the backward reaction rate. 1

(b) • K_c would increase. 1

• As the forward reaction is endothermic, an increase in temperature favours endothermic reaction, and the equilibrium position shifts to the right / product side. 1

10. (a) $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$ 1

(b) Manganese illustrates catalytic property because the rate of formation of gas bubbles increases when $MnO_2(s)$ is present. Moreover, MnO_2 remained chemically unchanged at the end of the reaction. 1

(c) Theoretical volume of $O_2(g)$ released $= 3.00 \times (10.0 \div 1000) \div 2 \times 24$ 2

$= 0.36$ dm^3

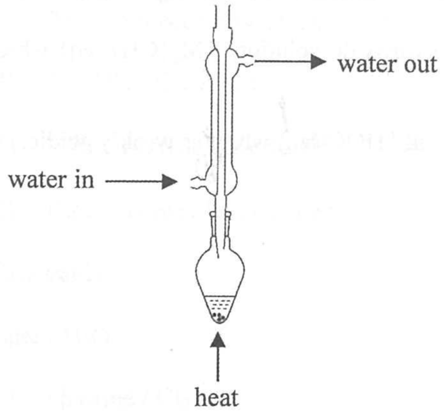
(d) • The concentration of reactant (H_2O_2) decreases in the progress of the reaction. 1

• The rate of decomposition of H_2O_2 decreases progressively. Thus, it takes longer time for the foam to reach every 100 cm^3–mark. 1

Marks

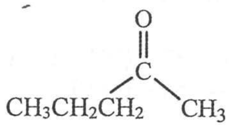
11. (a) 2-methylbutan-2-ol 1

(b) (i) 2

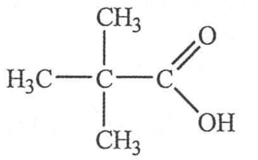


(ii) The reaction mixture changes from orange to green. 1

(iii) 1



(c) (i) 1



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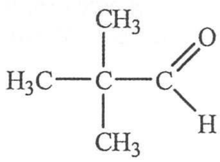
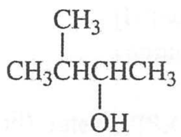
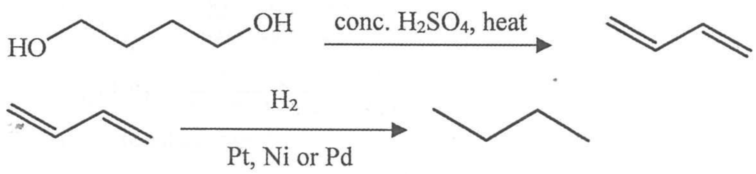


(ii) LiAlH$_4$ (if answered carboxylic acid in (i)) or LiAlH$_4$ / NaBH$_4$ (if answered aldehyde in (i)) 1

(d) 1



12. 3



Marks

13. Chemical knowledge 4

- $Na_2O(s)$ reacts with water to form sodium hydroxide solution / $NaOH(aq)$ which is strongly alkaline.
- $MgO(s)$ reacts with water to form magnesium hydroxide solution / $Mg(OH)_2(aq)$ which is weakly alkaline.
- $Al_2O_3(s)$ does not react with water.
- $Cl_2O(s)$ reacts with water to form hypochlorous acid / $HOCl(aq)$ which is weakly acidic.

Communication mark 1

Paper 2

Marks

1. (a) (i) (1) Catalysts (HI and Rh) are used. / Methanol can be made from renewable biomass. / The atom economy of the reaction is 100%. — 1

(2) CH_3OH / CO is toxic. — 1

(ii) (1) It can increase the effectiveness of the catalyst by increasing the surface area. — 1

(2) Catalysts may be poisoned by impurities. — 1

(iii) glass bottle — 1

(b) (i) water / H_2O — 1

(ii) (1) chlorine / Cl_2 — 1

(2) The concentration of $Cl^-(aq)$ ions is much higher than that of $OH^-(aq)$ ions, $Cl^-(aq)$ ions are preferentially discharged. — 1

(iii) (1) $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ / $2H^+(aq) + 2e^- \rightarrow H_2(g)$ — 1

(2)
- $OH^-(aq)$ ions are continuously formed at the cathode. — 1
- The ion-permeable membrane only allows $Na^+(aq)$ ions but not $Cl^-(aq)$ ions to pass through. — 1

(iv) sodium hypochlorite / NaOCl — 1

(c) (i) 'Initial rate' is the instantaneous rate at the start of a reaction. — 1

(ii)
- Since $[H^+(aq)]$ is much higher than $[S_2O_3^{2-}(aq)]$, there is relatively small change in $[H^+(aq)]$ in the reaction. — 1
- $[H^+(aq)]$ is regarded as a constant, then $k[H^+(aq)]^b$ can also be regarded as a constant. — 1

(iii) rate = $k'[S_2O_3^{2-}(aq)]^a$ — 3

log (rate) = log k' + $\boldsymbol{a}$ log ($[S_2O_3^{2-}(aq)]$)   ($\boldsymbol{a}$ = the slope of the straight line)

slope = [−1.10 − (−1.50)] ÷ [−1.84 − (−2.24)] = 1

The order of reaction with respect to $S_2O_3^{2-}(aq)$ is 1.

(iv) — 2

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.3R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log(1.9) = \frac{E_a}{2.3 \times 8.31}\left(\frac{1}{298} - \frac{1}{308}\right)$$

Ea = 48.90 kJ mol^{-1}

Marks

2. (a) (i) (1) Particles with sizes between 1–100 nm 1

(2) Making stained glass 1

(ii) (1) Diagram **C** 1

(2) nematic phase 1

(iii) Atom economy = $\frac{32.0}{2\times40.0+71.0+2\times17.0} \times 100\% = 17.3\%$ 1

(b) (i) face-centred cubic / cubic close-packed 1

(ii) Number of aluminium atoms in the unit cell = $8 \times \frac{1}{8} + 6 \times \frac{1}{2} = 4$ 1

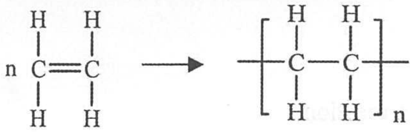
(iii) Density of aluminium = $27.0 \times 4 \div (6.02 \times 10^{23}) \div (4.05 \times 10^{-8})^3$ 2

$= 2.70 \text{ g cm}^{-3}$

(iv) (1)
- The addition of other metal atoms with different sizes distorts the regular arrangement of atoms in pure aluminium. 1
- This reduces the strength of metallic bonds. 1

(2) It is correct because this kind of duralumin has a higher tensile strength than pure aluminium. 1

(c) (i) 1



(ii) addition polymerisation 1

(iii) (1)
- HDPE chains are more linear and packed more closely than LDPE chains. 1
- There are stronger van der Waals' forces between HDPE chains than LDPE chains. 1

(2) blow moulding 1

(iv) (1) thermosetting 1

(2) Because there are strong cross-linkages between polymer chains. 1

(3) **X** has high hardness. 1

Marks

3. (a) (i)
- Pass SO_2(g) and CO_2(g) to acidified $K_2Cr_2O_7$(aq) separately. 1
- SO_2(g) can turn acidified $K_2Cr_2O_7$(aq) from orange to green, while there is no observable change for CO_2(g). 1

(ii)
- In the mass spectrum of CH_3CH_2CHO, a peak appears at m/z = 29 corresponding to CHO^+, while this peak does not appear in the mass spectrum of CH_3COCH_3. 1
- In the mass spectrum of CH_3COCH_3, a peak appears at m/z = 43 corresponding to CH_3CO^+, while this peak does not appear in the mass spectrum of CH_3CH_2CHO. 1

(iii) anhydrous sodium sulphate 1

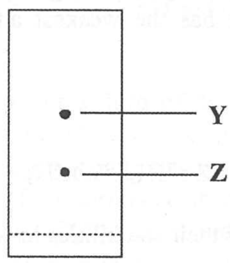
(b) (i) The maximum mass of **Y** dissolving in 50 cm^3 of water = $3.04 \times 50 \div 100 = 1.52$ g 1
The mass of **Y** in the solid sample (smaller than 1.40 g) is smaller than 1.52 g, thus all **Y** should have dissolved.

(ii) To remove the water-insoluble activated charcoal. 1

(iii) crystallisation 1

(iv) Some **Y** may not crystallise out. / Some **Y** solid may remain on the filter paper. / Some **Y** solid may be washed way. 1

(v) (1) 2

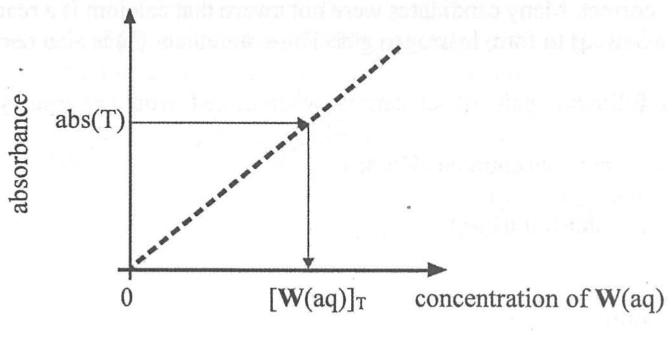


(2) **Y**, because the R_f value of **Y** is greater than that of **Z**, thus **Y** moves faster than **Z**. 1

(c) (i) (1)
- pale yellow colour : Fe^{3+}(aq) ions 1
- pale pink colour : MnO_4^-(aq) ions 1

(2) Concentration of Fe^{2+}(aq) ions in solution **S** = $0.0041 \times 32.35 \times 5 \div 25.00$ 2
= 0.0265 M

(ii) (1) 3



(2) Concentration of Fe^{2+}(aq) ions in solution **S** 1
= concentration of **W**(aq) in solution **T** × 100